

ZERO-TO-JOB-READY DATA ANALYST 2026

M09 - INTEGRATED BUSINESS ANALYSIS

Decision -> KPI -> diagnosis -> evidence -> recommendation -> communication -> defense

REQUEST -> DEFINITION -> VALIDATED EVIDENCE -> ACTION -> DECISION LOG

More charts never compensate for a broken evidence chain.

This book is your teacher

- M09 is not another Excel/SQL/Power BI/Python feature course. You may use any taught tool; decision quality and traceability are what count.
- Every lesson owns one direct worksheet in the rebuilt RoutePulse workbook, so you never have to guess where IBA3 ends or IBA4 begins.
- Every material number must trace to a metric definition, grain/source and independent validation. Weighted rates/totals must remain mathematically correct.
- Observational association is not causal proof. Recommendations describe owned tests/actions and measurable success; expected benefits are hypotheses.
- Videos are used where visual communication genuinely helps. Book-led lessons are not missing content—the internal explanation is complete.

Normal M09 loop

1. State decision first
2. Open exact IBA worksheet
3. Define metric/grain before calculation
4. Watch mapped visual if useful
5. Build one artifact
6. Validate materially
7. Close help + changed retry
8. Defend + append evidence

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Module prerequisite and tool map

0. Prerequisite

M00-M08 competency: framing, grain, KPI/rate definitions, data quality, Excel/SQL/Power BI/Python analysis and statistics/causal caution. M09 integrates those skills; it does not require a new technical tool.

Files / tools

- C1_M09_BUSINESS_ANALYSIS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx - controlled RoutePulse source + one direct IBA1-IBA11 sheet.
- C1_M09_BUSINESS_ANALYSIS_PROTECTED_PRACTICE_REVIEW_RC1.pdf - open only after saved attempt.
- C1_M09_BUSINESS_ANALYSIS_TARGETED_REPAIR_CARDS_RC1.pdf - localized repair route.
- Gate A CareBridge / Gate B StreamFit / Gate C QuickCart workbooks + instruction/review PDFs.
- Use one primary calculation tool and independently reconcile at least one material result.

One-to-one route repair

The inherited workbook reused generic Analysis_Plan for IBA3/4 and Evidence_Chain for IBA5/6.

RC1 keeps the same competencies/data but gives each lesson its own worksheet and fresh task.

This is structural regeneration, not a front-page patch.

Exact practice and Gate route

- IBA1 brief -> IBA2 KPI definitions -> IBA3 prioritized plan -> IBA4 diagnostic analysis -> IBA5 causal boundary -> IBA6 insight chain -> IBA7 recommendation -> IBA8 memo -> IBA9 story -> IBA10 defense -> IBA11 integrated case.
- Protected practice review opens only after a genuine saved artifact; after diagnosis close review and solve a changed task.
- IBA11 integrates the whole RoutePulse chain and is the readiness project before Gate A.
- Gate A CareBridge is normal. A clean pass ends M09.
- Gate B StreamFit and Gate C QuickCart are fresh remediation full retests only after prior failure + review + targeted repair + explicit assignment.

Critical failures

Fabricated evidence; silent denominator/grain error; hidden quality handling; causal overclaim;
recommendation not traceable to validated evidence; changing definition to fit desired story;
or inability to defend material numbers can block a pass.

IBA1 - Stakeholder request -> decision-focused Analysis Brief

Open exactly: IBA1_Analysis_Brief in C1_M09_BUSINESS_ANALYSIS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Decision quality depends on an auditable chain from request to evidence to action. More charts or more tool complexity do not compensate for a vague decision, wrong denominator, unreconciled output or unsupported causal claim.

2. What you are learning

Convert a vague stakeholder concern into a decision, analytical question, scope and Definition of Done.

3. What you already need / prerequisite

Prior modules provide the technical tools; no new tool is required.

4. Picture it in your head

A stakeholder request is raw material, not yet an analysis plan. Imagine translating 'something feels worse' into the exact decision someone must make, then the question/evidence needed to make it.

5. New words before we use them

Decision = action/choice stakeholder must make. Analytical question = answerable question that can change that decision. Scope = included time/entities. Definition of Done = evidence threshold/output needed. Assumption = unresolved rule treated provisionally.

6. Watch / visual source

BOOK-LED. No external video is required for this competency. The learner book + IBA1_Analysis_Brief are the complete teacher/practice route.

IBA1 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Analysis starts with the decision, not the columns. Separate the business concern from the analytical question; specify comparison, period, scope, metric meaning, exclusions, assumptions and what would count as decision-ready evidence. Teacher action: On the RoutePulse practice case, complete the 'Analysis_Brief' artifact. Use the lesson definition first, then calculate only the evidence needed for that artifact. Save one validation note before writing a conclusion.

8. What you should see / be able to say

A second analyst can state the decision, analytical question, scope, comparison and Definition of Done without asking what you meant.

9. Why it worked

The artifact begins with the decision/definition, uses only needed evidence, keeps grain/denominator/validation visible and leaves a traceable path to the recommendation. Tool complexity cannot substitute for this chain.

10. Common beginner mistakes

Starting with 'analyze delivery' instead of a decision; no comparison; no Definition of Done; silently inventing business definitions.

11. If you get stuck - recovery ladder

If the brief is vague, circle the verb in the stakeholder request. Ask: what decision will be made differently? Then rewrite the request as one decision and one analytical question. Concept -> exact sheet/metric/filter -> structural skeleton -> protected review only after a saved attempt -> close review and solve a changed retry.

IBA1 - Do alone, validate and master

12. Now you do a small version alone

Rewrite a fresh stakeholder request about rising delivery cost into a decision-focused brief without looking at the example. Include one open question you would ask before analysis.

13. Validate your result

Complete a changed Analysis_Brief task independently and defend it under one challenge. Pass only if the artifact is decision-focused, validated and no stronger than the evidence supports.

14. Teach it back in your own words

Without the lesson open, explain stakeholder request -> decision-focused analysis brief in 60-90 seconds. State one decision use, one validation control, one failure mode and the wording boundary you would defend to a stakeholder.

15. Protected review + changed fresh retry

Save the IBA1 worksheet/evidence first. If repair is needed, open the protected IBA1 review, diagnose only the weak competency, close it, then solve the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save the artifact, source/grain note, validation, limitation and short own-words defense. PASS -> continue. PARTIAL -> targeted repair + fresh retry. FAIL -> return only to the named lesson/competency.

IBA2 - KPI trees, metric definitions and driver logic

Open exactly: IBA2_KPI_Definitions in C1_M09_BUSINESS_ANALYSIS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Decision quality depends on an auditable chain from request to evidence to action. More charts or more tool complexity do not compensate for a vague decision, wrong denominator, unreconciled output or unsupported causal claim.

2. What you are learning

Define outcome and diagnostic metrics precisely enough that another analyst can reproduce them.

3. What you already need / prerequisite

Prerequisite: IBA1 evidence plus earlier technical analysis skills.

4. Picture it in your head

A KPI tree is a cause-and-driver map, not a dashboard menu. Outcome sits at the top; driver metrics sit below. Each metric has a definition contract so two analysts calculate the same thing.

5. New words before we use them

Outcome KPI = primary result metric. Driver metric = factor plausibly related to outcome. Numerator/denominator = components of a rate. Exclusion = row/entity intentionally outside metric. Definition authority = stakeholder/source that owns business meaning.

6. Watch / visual source

BOOK-LED. No external video is required for this competency. The learner book + IBA2_KPI_Definitions are the complete teacher/practice route.

IBA2 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

A KPI tree is a diagnostic map, not a causal diagram. Every rate needs numerator, denominator, scope and exclusions. Driver metrics help locate movement; they do not automatically explain why it moved. Teacher action: On the RoutePulse practice case, complete the `KPI\_Definitions` artifact. Use the lesson definition first, then calculate only the evidence needed for that artifact. Save one validation note before writing a conclusion.

8. What you should see / be able to say

Two analysts using your definitions would calculate the same numerator, denominator, exclusions and scope; weighted rates are not replaced by averages of subgroup rates.

9. Why it worked

The artifact begins with the decision/definition, uses only needed evidence, keeps grain/denominator/validation visible and leaves a traceable path to the recommendation. Tool complexity cannot substitute for this chain.

10. Common beginner mistakes

Metric name without numerator/denominator; averaging subgroup rates; drivers treated as proven causes; definition owner missing.

11. If you get stuck - recovery ladder

If KPI results disagree, stop analysis and audit definition, numerator, denominator, exclusions, time window and grain. Do not average rates unless weighting is mathematically valid. Concept -> exact sheet/metric/filter -> structural skeleton -> protected review only after a saved attempt -> close review and solve a changed retry.

IBA2 - Do alone, validate and master

12. Now you do a small version alone

Define a new rate KPI and one driver metric for a support operation. Write numerator, denominator, exclusions, grain and business owner.

13. Validate your result

Complete a changed KPI_Definitions task independently and defend it under one challenge. Pass only if the artifact is decision-focused, validated and no stronger than the evidence supports.

14. Teach it back in your own words

Without the lesson open, explain kpi trees, metric definitions and driver logic in 60-90 seconds. State one decision use, one validation control, one failure mode and the wording boundary you would defend to a stakeholder.

15. Protected review + changed fresh retry

Save the IBA2 worksheet/evidence first. If repair is needed, open the protected IBA2 review, diagnose only the weak competency, close it, then solve the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save the artifact, source/grain note, validation, limitation and short own-words defense. PASS -> continue. PARTIAL -> targeted repair + fresh retry. FAIL -> return only to the named lesson/competency.

IBA3 - Analysis planning and prioritization

Open exactly: IBA3_Analysis_Plan in C1_M09_BUSINESS_ANALYSIS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Decision quality depends on an auditable chain from request to evidence to action. More charts or more tool complexity do not compensate for a vague decision, wrong denominator, unreconciled output or unsupported causal claim.

2. What you are learning

Choose the smallest set of analyses that can materially change the decision.

3. What you already need / prerequisite

Prerequisite: IBA2 evidence plus earlier technical analysis skills.

4. Picture it in your head

Your analysis plan is a queue with limited time. Rank cuts by how likely they are to change the decision, not by how many filters/charts are available.

5. New words before we use them

Hypothesis = testable possible explanation. Prioritization = rank by decision value, not curiosity. Grain = what one row/result represents. Comparison = versus target/time/segment/reference. Stop rule = evidence after which extra cuts add little decision value.

6. Watch / visual source

BOOK-LED. No external video is required for this competency. The learner book + IBA3_Analysis_Plan are the complete teacher/practice route.

IBA3 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Prioritize questions by decision impact, evidence availability and reversibility. A hypothesis is something to test, not an answer you are trying to prove. Attach a validation check to every material output before you open the tool. Teacher action: On the RoutePulse practice case, complete the `Analysis\_Plan` artifact. Use the lesson definition first, then calculate only the evidence needed for that artifact. Save one validation note before writing a conclusion.

8. What you should see / be able to say

Every planned analysis has a decision reason and a validation method. Removing a low-value cut would not weaken the decision.

9. Why it worked

The artifact begins with the decision/definition, uses only needed evidence, keeps grain/denominator/validation visible and leaves a traceable path to the recommendation. Tool complexity cannot substitute for this chain.

10. Common beginner mistakes

Boil-the-ocean plan; priority based on easy chart rather than decision value; no grain/comparison; every hypothesis treated equally.

11. If you get stuck - recovery ladder

If the plan becomes a list of every possible slice, score each question on decision impact and evidence availability. Keep only the cuts that can change an action. Concept -> exact sheet/metric/filter -> structural skeleton -> protected review only after a saved attempt -> close review and solve a changed retry.

IBA3 - Do alone, validate and master

12. Now you do a small version alone

Given six possible cuts, rank the top three for a fresh retention decision and justify why the other three can wait.

13. Validate your result

Complete a changed Analysis_Plan task independently and defend it under one challenge. Pass only if the artifact is decision-focused, validated and no stronger than the evidence supports.

14. Teach it back in your own words

Without the lesson open, explain analysis planning and prioritization in 60-90 seconds. State one decision use, one validation control, one failure mode and the wording boundary you would defend to a stakeholder.

15. Protected review + changed fresh retry

Save the IBA3 worksheet/evidence first. If repair is needed, open the protected IBA3 review, diagnose only the weak competency, close it, then solve the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save the artifact, source/grain note, validation, limitation and short own-words defense. PASS -> continue. PARTIAL -> targeted repair + fresh retry. FAIL -> return only to the named lesson/competency.

IBA4 - Diagnostic analysis: comparison, trend, segment and variance

Open exactly: IBA4_Diagnostic in C1_M09_BUSINESS_ANALYSIS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Decision quality depends on an auditable chain from request to evidence to action. More charts or more tool complexity do not compensate for a vague decision, wrong denominator, unreconciled output or unsupported causal claim.

2. What you are learning

Find where a business gap is concentrated using comparison, trend, segment and variance without cherry-picking.

3. What you already need / prerequisite

Prerequisite: IBA3 evidence plus earlier technical analysis skills.

4. Picture it in your head

Diagnostic analysis is a flashlight: compare overall vs target, then time/segment/channel/category to find where the gap concentrates. Stop when a cut no longer changes the decision.

5. New words before we use them

Diagnostic = analysis locating where/when pattern concentrates. Weighted rate = recompute numerator/denominator totals rather than average subgroup percentages. Variance = actual minus target/reference. Material = large enough to affect decision.

6. Watch / visual source

ASSIGNED VISUAL DEMONSTRATION. The art of exploring and explaining data - storytelling with data | 02:38-28:20. Why: See exploration, comparison, simple explanatory charts and the shift from many cuts to a few decision-relevant patterns. Internal fallback: this lesson + IBA4_Diagnostic fully teach the competency.

[Open assigned visual](#)

IBA4 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Diagnose from whole -> comparison -> trend -> segment -> variance. Compare like-for-like periods and denominators. Segment only where different findings could lead to different actions, and reconcile segment totals to the whole. Teacher action: On the RoutePulse practice case, complete the 'Analysis_Plan' artifact. Use the lesson definition first, then calculate only the evidence needed for that artifact. Save one validation note before writing a conclusion.

8. What you should see / be able to say

Segment totals reconcile to the whole, periods/denominators are comparable, and the priority gap is material rather than merely visually dramatic.

9. Why it worked

The artifact begins with the decision/definition, uses only needed evidence, keeps grain/denominator/validation visible and leaves a traceable path to the recommendation. Tool complexity cannot substitute for this chain.

10. Common beginner mistakes

Averaging subgroup percentages instead of weighting; comparing mismatched scopes; presenting a tiny difference as material; uncontrolled repeated slicing until a dramatic result appears.

11. If you get stuck - recovery ladder

If segments do not reconcile, check filters, denominator/grain, duplicate rows and time windows before interpreting. If one segment looks extreme, compare its size and persistence before prioritizing. Concept -> exact sheet/metric/filter -> structural skeleton -> protected review only after a saved attempt -> close review and solve a changed retry.

IBA4 - Do alone, validate and master

12. Now you do a small version alone

Use a changed slice of the practice data to identify whether the problem is broad or concentrated. Reconcile your selected segment to the total.

13. Validate your result

Complete a changed Analysis_Plan task independently and defend it under one challenge. Pass only if the artifact is decision-focused, validated and no stronger than the evidence supports.

14. Teach it back in your own words

Without the lesson open, explain diagnostic analysis: comparison, trend, segment and variance in 60-90 seconds. State one decision use, one validation control, one failure mode and the wording boundary you would defend to a stakeholder.

15. Protected review + changed fresh retry

Save the IBA4 worksheet/evidence first. If repair is needed, open the protected IBA4 review, diagnose only the weak competency, close it, then solve the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save the artifact, source/grain note, validation, limitation and short own-words defense. PASS -> continue. PARTIAL -> targeted repair + fresh retry. FAIL -> return only to the named lesson/competency.

IBA5 - Root-cause discipline and causal boundaries

Open exactly: IBA5_Causal_Boundary in C1_M09_BUSINESS_ANALYSIS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Decision quality depends on an auditable chain from request to evidence to action. More charts or more tool complexity do not compensate for a vague decision, wrong denominator, unreconciled output or unsupported causal claim.

2. What you are learning

Narrow plausible causes without turning observational association into causal proof.

3. What you already need / prerequisite

Prerequisite: IBA4 evidence plus earlier technical analysis skills.

4. Picture it in your head

Root-cause reasoning is a courtroom discipline. Observational data can show an association, but multiple causal stories can fit the same pattern. Label what is observed, hypothesized and unproven.

5. New words before we use them

Association = variables move together. Causal claim = says changing X produces Y. Confounder = third factor related to both. Reverse causality = Y may influence X. Stronger design = experiment/quasi-experiment or richer causal evidence.

6. Watch / visual source

BOOK-LED. No external video is required for this competency. The learner book + IBA5_Causal_Boundary are the complete teacher/practice route.

IBA5 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Use claim levels deliberately: descriptive pattern -> suggestive association -> causal evidence. Observational comparisons can narrow hypotheses, but strong causal wording requires a design that addresses alternative explanations. Teacher action: On the RoutePulse practice case, complete the `Evidence\_Chain` artifact. Use the lesson definition first, then calculate only the evidence needed for that artifact. Save one validation note before writing a conclusion.

8. What you should see / be able to say

Every causal word is supported by the evidence design. Association is labelled as association; competing explanations and missing data are visible.

9. Why it worked

The artifact begins with the decision/definition, uses only needed evidence, keeps grain/denominator/validation visible and leaves a traceable path to the recommendation. Tool complexity cannot substitute for this chain.

10. Common beginner mistakes

Calling association causal; one plausible story presented as fact; confounders ignored; 'root cause' language without stronger evidence.

11. If you get stuck - recovery ladder

If you catch yourself writing because, caused, drives or leads to, rewrite the claim at the evidence level you actually have and list at least one plausible confounder or alternative mechanism. Concept -> exact sheet/metric/filter -> structural skeleton -> protected review only after a saved attempt -> close review and solve a changed retry.

IBA5 - Do alone, validate and master

12. Now you do a small version alone

Take a fresh correlation/association in the practice data and write one descriptive claim, one cautious interpretation and one causal claim you must NOT make.

13. Validate your result

Complete a changed Evidence_Chain task independently and defend it under one challenge. Pass only if the artifact is decision-focused, validated and no stronger than the evidence supports.

14. Teach it back in your own words

Without the lesson open, explain root-cause discipline and causal boundaries in 60-90 seconds. State one decision use, one validation control, one failure mode and the wording boundary you would defend to a stakeholder.

15. Protected review + changed fresh retry

Save the IBA5 worksheet/evidence first. If repair is needed, open the protected IBA5 review, diagnose only the weak competency, close it, then solve the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save the artifact, source/grain note, validation, limitation and short own-words defense. PASS -> continue. PARTIAL -> targeted repair + fresh retry. FAIL -> return only to the named lesson/competency.

IBA6 - Observation -> insight -> implication

Open exactly: IBA6_Insight_Chain in C1_M09_BUSINESS_ANALYSIS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Decision quality depends on an auditable chain from request to evidence to action. More charts or more tool complexity do not compensate for a vague decision, wrong denominator, unreconciled output or unsupported causal claim.

2. What you are learning

Synthesize facts into decision-relevant insights with comparison, magnitude, scope and implication.

3. What you already need / prerequisite

Prerequisite: IBA5 evidence plus earlier technical analysis skills.

4. Picture it in your head

Observation says what happened. Insight adds comparison, magnitude and scope. Implication explains why that matters to the decision. Each arrow must be supported.

5. New words before we use them

Observation = factual output. Insight = decision-relevant interpretation with comparison/magnitude/scope. Implication = consequence for choice/action. Materiality = whether difference is large enough to matter.

6. Watch / visual source

ASSIGNED VISUAL DEMONSTRATION. The art of exploring and explaining data - storytelling with data | 17:15-28:20. Why: See how exploratory output becomes a focused explanatory message. Internal fallback: this lesson + IBA6_Insight_Chain fully teach the competency.

[Open assigned visual](#)

IBA6 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Observation says what happened. Insight says why the pattern matters in context. Implication says what the evidence means for the decision. Strong insights include comparison, magnitude, scope and a causal boundary. Teacher action: On the RoutePulse practice case, complete the `Evidence\_Chain` artifact. Use the lesson definition first, then calculate only the evidence needed for that artifact. Save one validation note before writing a conclusion.

8. What you should see / be able to say

Each insight contains comparison, magnitude, scope and why it matters. It does not simply rename a chart or restate a number.

9. Why it worked

The artifact begins with the decision/definition, uses only needed evidence, keeps grain/denominator/validation visible and leaves a traceable path to the recommendation. Tool complexity cannot substitute for this chain.

10. Common beginner mistakes

Insight is only a restated number; no versus-what/magnitude/scope; implication not linked to decision; non-material pattern overstated.

11. If you get stuck - recovery ladder

If an insight is just a metric, add versus what, by how much, where/when and what decision it changes. If you cannot add those, you may only have an observation. Concept -> exact sheet/metric/filter -> structural skeleton -> protected review only after a saved attempt -> close review and solve a changed retry.

IBA6 - Do alone, validate and master

12. Now you do a small version alone

Convert three fresh outputs into observation -> insight -> implication statements. At least one should conclude that the pattern is not material enough to change the decision.

13. Validate your result

Complete a changed Evidence_Chain task independently and defend it under one challenge. Pass only if the artifact is decision-focused, validated and no stronger than the evidence supports.

14. Teach it back in your own words

Without the lesson open, explain observation -> insight -> implication in 60-90 seconds. State one decision use, one validation control, one failure mode and the wording boundary you would defend to a stakeholder.

15. Protected review + changed fresh retry

Save the IBA6 worksheet/evidence first. If repair is needed, open the protected IBA6 review, diagnose only the weak competency, close it, then solve the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save the artifact, source/grain note, validation, limitation and short own-words defense. PASS -> continue. PARTIAL -> targeted repair + fresh retry. FAIL -> return only to the named lesson/competency.

IBA7 - Recommendations that can actually be executed

Open exactly: IBA7_Recommendation in C1_M09_BUSINESS_ANALYSIS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Decision quality depends on an auditable chain from request to evidence to action. More charts or more tool complexity do not compensate for a vague decision, wrong denominator, unreconciled output or unsupported causal claim.

2. What you are learning

Turn evidence into an owned action, success measure, risk and next validation.

3. What you already need / prerequisite

Prerequisite: IBA6 evidence plus earlier technical analysis skills.

4. Picture it in your head

A recommendation is a testable operating instruction: owner + action + scope/timing + success measure + risk + stop/continue rule. It is not 'improve performance'.

5. New words before we use them

Owner = accountable role. Success measure = metric that judges test. Dependency = condition needed. Stop/continue rule = predeclared decision threshold. Expected benefit = hypothesis, not guaranteed outcome.

6. Watch / visual source

BOOK-LED. No external video is required for this competency. The learner book + IBA7_Recommendation are the complete teacher/practice route.

IBA7 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

A recommendation is not advice such as improve operations. It is owner + action + scope + timing + success measure + risk/dependency + next validation. Expected benefit is a measurable hypothesis, not a promise. Teacher action: On the RoutePulse practice case, complete the `Recommendation\_Log` artifact. Use the lesson definition first, then calculate only the evidence needed for that artifact. Save one validation note before writing a conclusion.

8. What you should see / be able to say

The named owner could start tomorrow and knows what success means, what could go wrong and when the recommendation should be reconsidered.

9. Why it worked

The artifact begins with the decision/definition, uses only needed evidence, keeps grain/denominator/validation visible and leaves a traceable path to the recommendation. Tool complexity cannot substitute for this chain.

10. Common beginner mistakes

Generic 'improve X'; no owner/timing/success metric; expected benefit promised as fact; no stop condition.

11. If you get stuck - recovery ladder

If the recommendation is generic, force five fields: owner, first action, scope/time, success metric and risk/stop condition. Remove any promised benefit that the evidence cannot support. Concept -> exact sheet/metric/filter -> structural skeleton -> protected review only after a saved attempt -> close review and solve a changed retry.

IBA7 - Do alone, validate and master

12. Now you do a small version alone

Write a recommendation for a different business owner with a 4-week test, success metric, risk and stop/continue rule.

13. Validate your result

Complete a changed Recommendation_Log task independently and defend it under one challenge. Pass only if the artifact is decision-focused, validated and no stronger than the evidence supports.

14. Teach it back in your own words

Without the lesson open, explain recommendations that can actually be executed in 60-90 seconds. State one decision use, one validation control, one failure mode and the wording boundary you would defend to a stakeholder.

15. Protected review + changed fresh retry

Save the IBA7 worksheet/evidence first. If repair is needed, open the protected IBA7 review, diagnose only the weak competency, close it, then solve the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save the artifact, source/grain note, validation, limitation and short own-words defense. PASS -> continue. PARTIAL -> targeted repair + fresh retry. FAIL -> return only to the named lesson/competency.

IBA8 - Executive writing: headline, evidence and one-page memo

Open exactly: IBA8_Executive_Memo in C1_M09_BUSINESS_ANALYSIS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Decision quality depends on an auditable chain from request to evidence to action. More charts or more tool complexity do not compensate for a vague decision, wrong denominator, unreconciled output or unsupported causal claim.

2. What you are learning

Write for a busy decision-maker: lead with the message, support it with the few strongest evidence points and state material limitations.

3. What you already need / prerequisite

Prerequisite: IBA7 evidence plus earlier technical analysis skills.

4. Picture it in your head

An executive memo is an inverted pyramid. Put the decision/message first, then the few strongest evidence points, then material limitations/next measurement.

5. New words before we use them

Message headline = sentence stating main conclusion. Evidence point = validated fact supporting it. Backup = detail not needed in first read. Material limitation = uncertainty that could change interpretation/action.

6. Watch / visual source

BOOK-LED. No external video is required for this competency. The learner book + IBA8_Executive_Memo are the complete teacher/practice route.

IBA8 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Executive communication leads with the decision-relevant message. Use 2-4 strongest evidence points, put method detail in backup unless it changes interpretation, and state limitations that constrain the action. Teacher action: On the RoutePulse practice case, complete the `Executive\_Memo` artifact. Use the lesson definition first, then calculate only the evidence needed for that artifact. Save one validation note before writing a conclusion.

8. What you should see / be able to say

The first three sentences are useful on their own; evidence supports the headline; methodology does not crowd out the decision; limitations are proportionate.

9. Why it worked

The artifact begins with the decision/definition, uses only needed evidence, keeps grain/denominator/validation visible and leaves a traceable path to the recommendation. Tool complexity cannot substitute for this chain.

10. Common beginner mistakes

Methodology before message; too many evidence points; headline is a topic label; limitation buried or omitted.

11. If you get stuck - recovery ladder

If the memo starts with method or is longer than the decision needs, rewrite the headline as a complete message sentence, keep 2-4 evidence bullets and move detail to backup. Concept -> exact sheet/metric/filter -> structural skeleton -> protected review only after a saved attempt -> close review and solve a changed retry.

IBA8 - Do alone, validate and master

12. Now you do a small version alone

Write a 120-word decision memo for a different audience (finance vs operations) using the same evidence but changing emphasis appropriately.

13. Validate your result

Complete a changed Executive_Memo task independently and defend it under one challenge. Pass only if the artifact is decision-focused, validated and no stronger than the evidence supports.

14. Teach it back in your own words

Without the lesson open, explain executive writing: headline, evidence and one-page memo in 60-90 seconds. State one decision use, one validation control, one failure mode and the wording boundary you would defend to a stakeholder.

15. Protected review + changed fresh retry

Save the IBA8 worksheet/evidence first. If repair is needed, open the protected IBA8 review, diagnose only the weak competency, close it, then solve the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save the artifact, source/grain note, validation, limitation and short own-words defense. PASS -> continue. PARTIAL -> targeted repair + fresh retry. FAIL -> return only to the named lesson/competency.

IBA9 - Data storytelling and visual hierarchy

Open exactly: IBA9_Story_Plan in C1_M09_BUSINESS_ANALYSIS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Decision quality depends on an auditable chain from request to evidence to action. More charts or more tool complexity do not compensate for a vague decision, wrong denominator, unreconciled output or unsupported causal claim.

2. What you are learning

Turn exploratory outputs into a focused communication sequence that guides one decision.

3. What you already need / prerequisite

Prerequisite: IBA8 evidence plus earlier technical analysis skills.

4. Picture it in your head

A data story is a guided path through evidence. Each slide/page has one message headline; position, annotation and contrast tell the audience where to look first.

5. New words before we use them

Message headline = conclusion, not topic label. Visual hierarchy = order audience notices elements. Annotation = text pointing to evidence. Clutter = visual/detail not helping decision. Storyboard = ordered message plan.

6. Watch / visual source

ASSIGNED VISUAL DEMONSTRATION. How to turn data into stories - storytelling with data | 02:20-20:52; 25:54-35:38. Why: Study decluttering, focus and a decision-ready story sequence. Internal fallback: this lesson + IBA9_Story_Plan fully teach the competency. Optional companions: Avoid these 5 slide mistakes, How to focus attention in data visualization.

[Open assigned visual](#)

[Open optional companion](#)

IBA9 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Exploration and explanation are different jobs. A story removes outputs that do not serve the decision, uses message headlines, directs attention with position/annotation/contrast and gives each slide/page one main point. Teacher action: On the RoutePulse practice case, complete the `Visual\_Story\_Plan` artifact. Use the lesson definition first, then calculate only the evidence needed for that artifact. Save one validation note before writing a conclusion.

8. What you should see / be able to say

The audience can identify the main message of each page/slide within seconds. Every visual earns its place and reconciles to the validated analysis.

9. Why it worked

The artifact begins with the decision/definition, uses only needed evidence, keeps grain/denominator/validation visible and leaves a traceable path to the recommendation. Tool complexity cannot substitute for this chain.

10. Common beginner mistakes

Dashboard screenshot as story; too many colors/graphs; title names metric rather than message; visual does not reconcile to validated analysis.

11. If you get stuck - recovery ladder

If the story looks like a dashboard screenshot, write the decision sequence first, then choose one visual per message. Remove decorative elements and annotate the evidence that matters. Concept -> exact sheet/metric/filter -> structural skeleton -> protected review only after a saved attempt -> close review and solve a changed retry.

IBA9 - Do alone, validate and master

12. Now you do a small version alone

Turn four fresh findings into a 3-slide story: decision/problem, concentrated evidence, recommendation/measurement. Give each slide a message headline.

13. Validate your result

Complete a changed Visual_Story_Plan task independently and defend it under one challenge. Pass only if the artifact is decision-focused, validated and no stronger than the evidence supports.

14. Teach it back in your own words

Without the lesson open, explain data storytelling and visual hierarchy in 60-90 seconds. State one decision use, one validation control, one failure mode and the wording boundary you would defend to a stakeholder.

15. Protected review + changed fresh retry

Save the IBA9 worksheet/evidence first. If repair is needed, open the protected IBA9 review, diagnose only the weak competency, close it, then solve the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save the artifact, source/grain note, validation, limitation and short own-words defense. PASS -> continue. PARTIAL -> targeted repair + fresh retry. FAIL -> return only to the named lesson/competency.

IBA10 - Stakeholder defense: questions, pushback and uncertainty

Open exactly: IBA10_Defense in C1_M09_BUSINESS_ANALYSIS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Decision quality depends on an auditable chain from request to evidence to action. More charts or more tool complexity do not compensate for a vague decision, wrong denominator, unreconciled output or unsupported causal claim.

2. What you are learning

Defend definitions, evidence and limitations without bluffing or overclaiming.

3. What you already need / prerequisite

Prerequisite: IBA9 evidence plus earlier technical analysis skills.

4. Picture it in your head

Stakeholder defense is not winning an argument. Answer first, reproduce definitions/evidence, separate known from uncertain, and say what additional evidence would resolve the uncertainty.

5. New words before we use them

Definition dispute = disagreement about business meaning. Calculation error = implementation mistake. Known = supported. Uncertain = plausible but unresolved. Unsupported = data cannot justify it. Challenge log = evidence trail of question/resolution.

6. Watch / visual source

BOOK-LED. No external video is required for this competency. The learner book + IBA10_Defense are the complete teacher/practice route.

IBA10 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

When challenged, answer the question first, then show evidence. Separate a definition dispute from a calculation error. Reproduce the logic; say I do not know yet when the data cannot support a stronger answer. Teacher action: On the RoutePulse practice case, complete the `Validation\_Log` artifact. Use the lesson definition first, then calculate only the evidence needed for that artifact. Save one validation note before writing a conclusion.

8. What you should see / be able to say

You can reproduce the challenged definition/filter/reconciliation and distinguish known, uncertain and unsupported claims without changing the story to win the argument.

9. Why it worked

The artifact begins with the decision/definition, uses only needed evidence, keeps grain/denominator/validation visible and leaves a traceable path to the recommendation. Tool complexity cannot substitute for this chain.

10. Common beginner mistakes

Bluffing; changing metric definition under pressure; confusing uncertainty with error; answering a causal challenge with correlation.

11. If you get stuck - recovery ladder

If a stakeholder challenge exposes a real definition/calculation error, correct it and update the decision log. If it exposes uncertainty, state what additional evidence would resolve it. Concept -> exact sheet/metric/filter -> structural skeleton -> protected review only after a saved attempt -> close review and solve a changed retry.

IBA10 - Do alone, validate and master

12. Now you do a small version alone

Role-play three challenges: denominator dispute, causal pushback and missing-data question. Answer each in 3-5 sentences with evidence and uncertainty.

13. Validate your result

Complete a changed Validation_Log task independently and defend it under one challenge. Pass only if the artifact is decision-focused, validated and no stronger than the evidence supports.

14. Teach it back in your own words

Without the lesson open, explain stakeholder defense: questions, pushback and uncertainty in 60-90 seconds. State one decision use, one validation control, one failure mode and the wording boundary you would defend to a stakeholder.

15. Protected review + changed fresh retry

Save the IBA10 worksheet/evidence first. If repair is needed, open the protected IBA10 review, diagnose only the weak competency, close it, then solve the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save the artifact, source/grain note, validation, limitation and short own-words defense. PASS -> continue. PARTIAL -> targeted repair + fresh retry. FAIL -> return only to the named lesson/competency.

IBA11 - Integrated case: analysis -> decision -> communication

Open exactly: IBA11_Integrated_Case in C1_M09_BUSINESS_ANALYSIS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Decision quality depends on an auditable chain from request to evidence to action. More charts or more tool complexity do not compensate for a vague decision, wrong denominator, unreconciled output or unsupported causal claim.

2. What you are learning

Execute the complete workflow on an unfamiliar business problem and leave a traceable evidence chain.

3. What you already need / prerequisite

Prerequisite: IBA10 evidence plus earlier technical analysis skills.

4. Picture it in your head

Integrated analysis is a chain of custody: recommendation -> implication -> insight -> observation -> validated metric -> definition -> grain/source. A broken upstream link invalidates downstream polish.

5. New words before we use them

Evidence chain = traceable dependency from source to decision. Decision log = record of action/evidence/assumptions/owner/review. Reconciliation = independent validation. Handoff = package another analyst/stakeholder can audit.

6. Watch / visual source

OPTIONAL VISUAL COMPANION. The art of exploring and explaining data - persuasion/story section - storytelling with data | 32:30-43:46. Why: Use after the integrated case draft to strengthen persuasion/story flow without replacing the evidence chain. Internal fallback: this lesson + IBA11_Integrated_Case fully teach the competency.

[Open assigned visual](#)

IBA11 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

The finished workflow is request -> decision -> question -> metric definition -> data/grain -> validation -> analysis -> observation -> insight -> implication -> recommendation -> owner/success/risk -> communication -> decision log. Teacher action: On the RoutePulse practice case, complete the `Integrated\_Case` artifact. Use the lesson definition first, then calculate only the evidence needed for that artifact. Save one validation note before writing a conclusion.

8. What you should see / be able to say

A reviewer can trace every recommendation backward to validated evidence and every material number back to a definition, grain and validation record.

9. Why it worked

The artifact begins with the decision/definition, uses only needed evidence, keeps grain/denominator/validation visible and leaves a traceable path to the recommendation. Tool complexity cannot substitute for this chain.

10. Common beginner mistakes

Broken evidence chain hidden by polished slides; recommendation has no traceable metric; contradictory definitions across artifacts; validation performed too late.

11. If you get stuck - recovery ladder

If the case feels overwhelming, return to the evidence chain and find the first broken arrow. Repair that dependency rather than polishing downstream slides. Concept -> exact sheet/metric/filter -> structural skeleton -> protected review only after a saved attempt -> close review and solve a changed retry.

IBA11 - Do alone, validate and master

12. Now you do a small version alone

Complete a changed integrated mini-case with a new decision and produce a brief, KPI definitions, validation evidence, three findings, one recommendation and a 3-slide plan.

13. Validate your result

Complete a changed Integrated_Case task independently and defend it under one challenge. Pass only if the artifact is decision-focused, validated and no stronger than the evidence supports.

14. Teach it back in your own words

Without the lesson open, explain integrated case: analysis -> decision -> communication in 60-90 seconds. State one decision use, one validation control, one failure mode and the wording boundary you would defend to a stakeholder.

15. Protected review + changed fresh retry

Save the IBA11 worksheet/evidence first. If repair is needed, open the protected IBA11 review, diagnose only the weak competency, close it, then solve the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save the artifact, source/grain note, validation, limitation and short own-words defense. PASS -> continue. PARTIAL -> targeted repair + fresh retry. FAIL -> return only to the named lesson/competency.

End-of-module mastery checklist

- IBA1: Decision-focused brief with question, scope, comparison, DoD and open assumptions.
- IBA2: Outcome/driver KPIs with numerator, denominator, exclusions, grain, owner and validation.
- IBA3: Top analyses prioritized by decision value rather than curiosity.
- IBA4: Diagnostic cuts use comparable scopes, weighted rates/totals and independent reconciliation.
- IBA5: Observational association separated from causal hypothesis/proof.
- IBA6: Insights contain comparison, magnitude, scope and decision implication.
- IBA7: Recommendation has owner, action, timing/scope, success measure, risk and stop/continue rule.
- IBA8: Executive memo leads with message and strongest evidence; material limitation visible.
- IBA9: Three-slide story has message headlines and purposeful visual hierarchy.
- IBA10: Stakeholder challenges answered with evidence, definitions and uncertainty—not bluffing.
- IBA11: Reviewer can trace recommendation backward to validated source/grain/definition.

Gate readiness

Attempt Gate A only when IBA11 is complete and one material result is independently reconciled.

A clean Gate A pass ends M09; B/C are remediation full retests only.

Revision / business-analysis quick reference

Decision first	What choice will this analysis change?
Metric contract	Meaning + numerator/denominator + exclusions + grain + owner.
Prioritize	Analyze cuts likely to change the decision first.
Diagnose	Overall -> target/time/segment/channel/category; keep scopes comparable.
Validate	Reconcile one material result independently; weighted rates/totals.
Causal boundary	Association supports description/hypothesis, not causal proof.
Insight	Observation + comparison + magnitude + scope + why it matters.
Recommendation	Owner + action + timing/scope + success + risk + stop/continue.
Executive memo	Message -> 2-4 strongest evidence -> limitation -> next measurement.
Story	One message headline per page; focus audience on decision evidence.
Defense	Answer first; reproduce evidence; say what is uncertain/unsupported.

Next route: IBA11 -> Gate A -> M10 AI-Assisted Analytics after M09 competency pass.